

A Computational Examination of Virtue Ethics Presence in Three Textual Datasets

Evelin Buestan '26, Lucy Engels '27, Artemist Pierre '27, Yuhe Xue '27
Advised by Prof. Eni Mustafaraj (Computer Science)

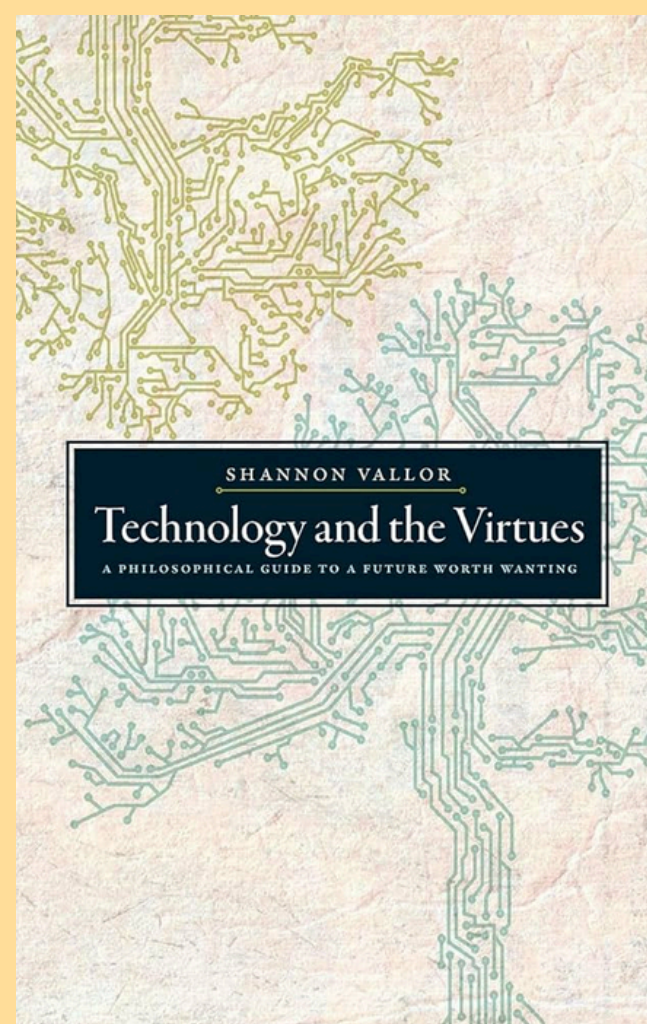


Background

Ethics of technology is a critical area of study as its impacts are reflected on both individuals and societies through the use of technologies. More broadly, people have enriched their lives and communities by following ethical principles. In her book *Technology and the Virtues*, Shannon Vallor outlined 12 techno-moral virtues and talked about the importance of practicing those virtues as both users and builders of technologies. Our research asks this question: **to what extent can we observe evidence of presence or engagement with Vallor's virtues in the textual sources of Wellesley's curriculum, the Wellesley News, and the and personal communications on the anonymous messaging board Sidechat?** These sources were then scored and analyzed by both humans and GPT. The validation of the GPT analysis can be found in the companion poster, *An Assessment of GPT-Prompting for Labeling Text in the Context of Virtue Ethics*.

The Technomoral Virtues

1. **Honesty**
2. **Self-control**
3. **Humility**
4. **Justice**
5. **Courage**
6. **Empathy**
7. **Care**
8. **Civility**
9. **Flexibility**
10. **Perspective**
11. **Magnanimity**
12. **Technomoral wisdom**



Shannon Vallor, *Technology and the Virtues*

Summary of the Datasets

Course Catalog: **1815 courses** were collected from the Wellesley Course catalog. They were **evaluated for the extent they touch upon Shannon Vallor's technomoral virtues**. A score of 5 indicates that the virtue is minimally present and a score of 1 indicates that the virtue is very significantly present.

Wellesley News: **3504 articles** from 2013 to 2022 were collected and then measured on a 5-point scale for the their **relevance to the dimensions of "technology and its use" and "ethics and societal impact"**. A score of 1 means "minimally mentions", and a score of 5 means "heavily mentions".

SideChat: **2265 posts** were collected and **measured** on a 3-point scale, which was used to measure two dimensions, **Tone** and **Content**. The Tone is scored from -1 to 1, where -1 means being **rude** and 1 means being **polite**. The Content also uses the same scale, where -1 means the **opposite of Shannon Vallor's technomoral virtues** and 1 means **displaying the virtue fully**. 0 means neutral for both dimensions.

Wellesley Course Catalog

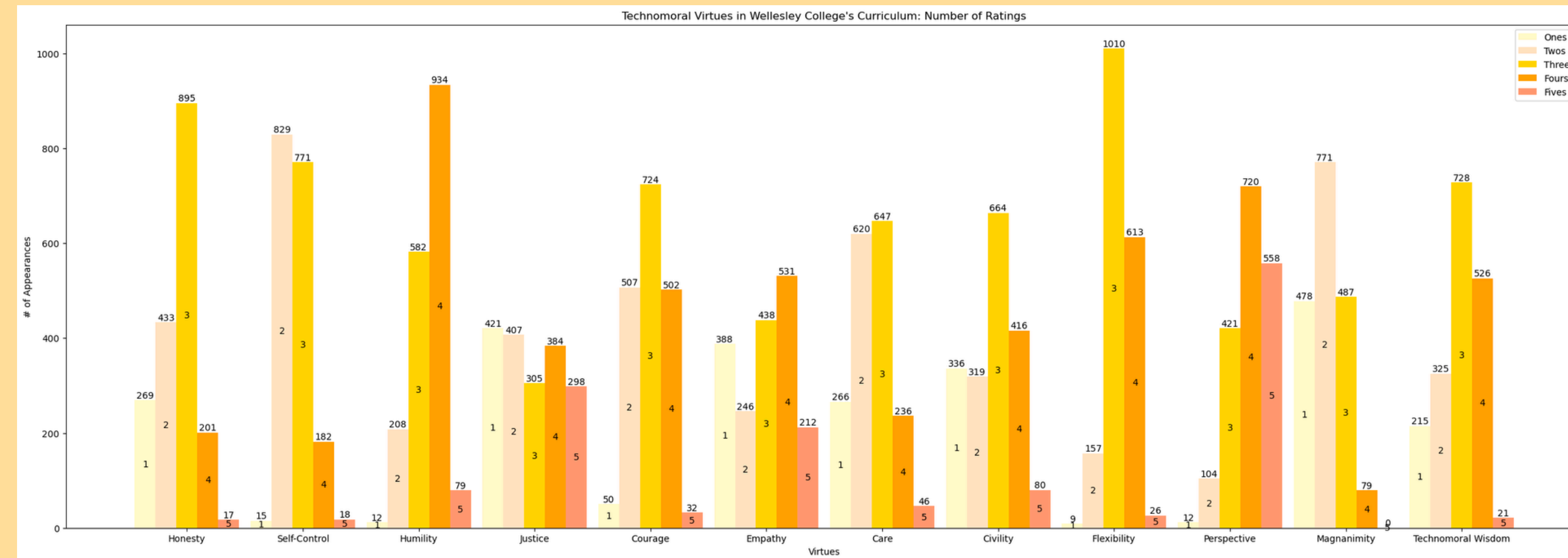


Figure 1. Bargraph for the distribution of scores according to each virtue

Perspective (558), **Justice (298)**, and **Empathy (212)** are the virtues that most frequently scored a five. **Magnanimity (0)**, **Honesty (17)**, and **Self-Control (18)** were the virtues that least frequently scored a five

708 (40%) Courses that recieved at least one score of 5

390 (21%) Courses that recieved at least two scores of 5

2.92 The mean score (the mean across all virtues in a course)

8 The greatest number of virtues scoring a 5 in one course. The one course that had a score of five on 8 virtues was PEAC / POL3 392: Truth Commissions and Conceptual Studies.

Wellesley News

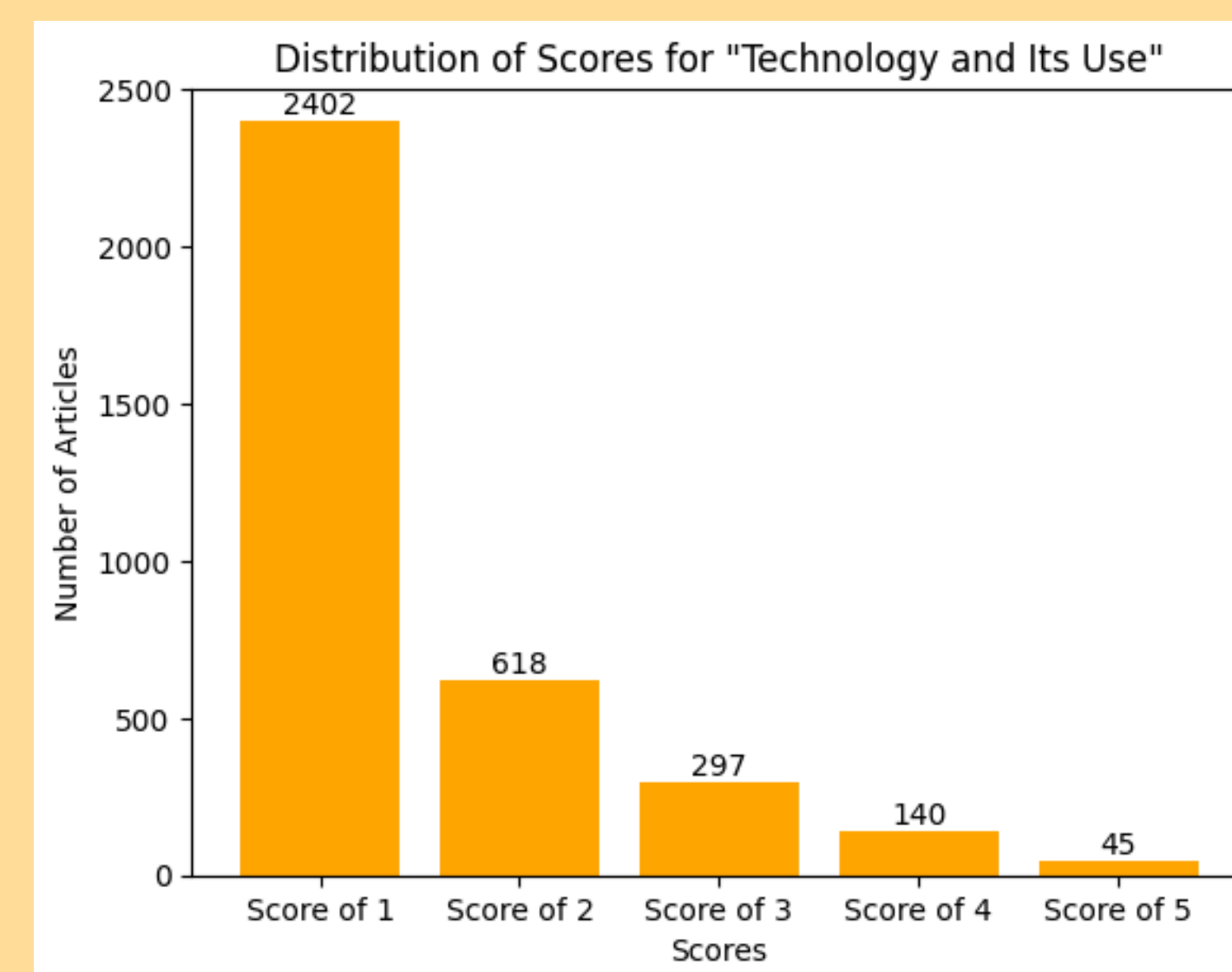


Figure 2. Bargraph for the distribution of scores for "Technology and Its Use"

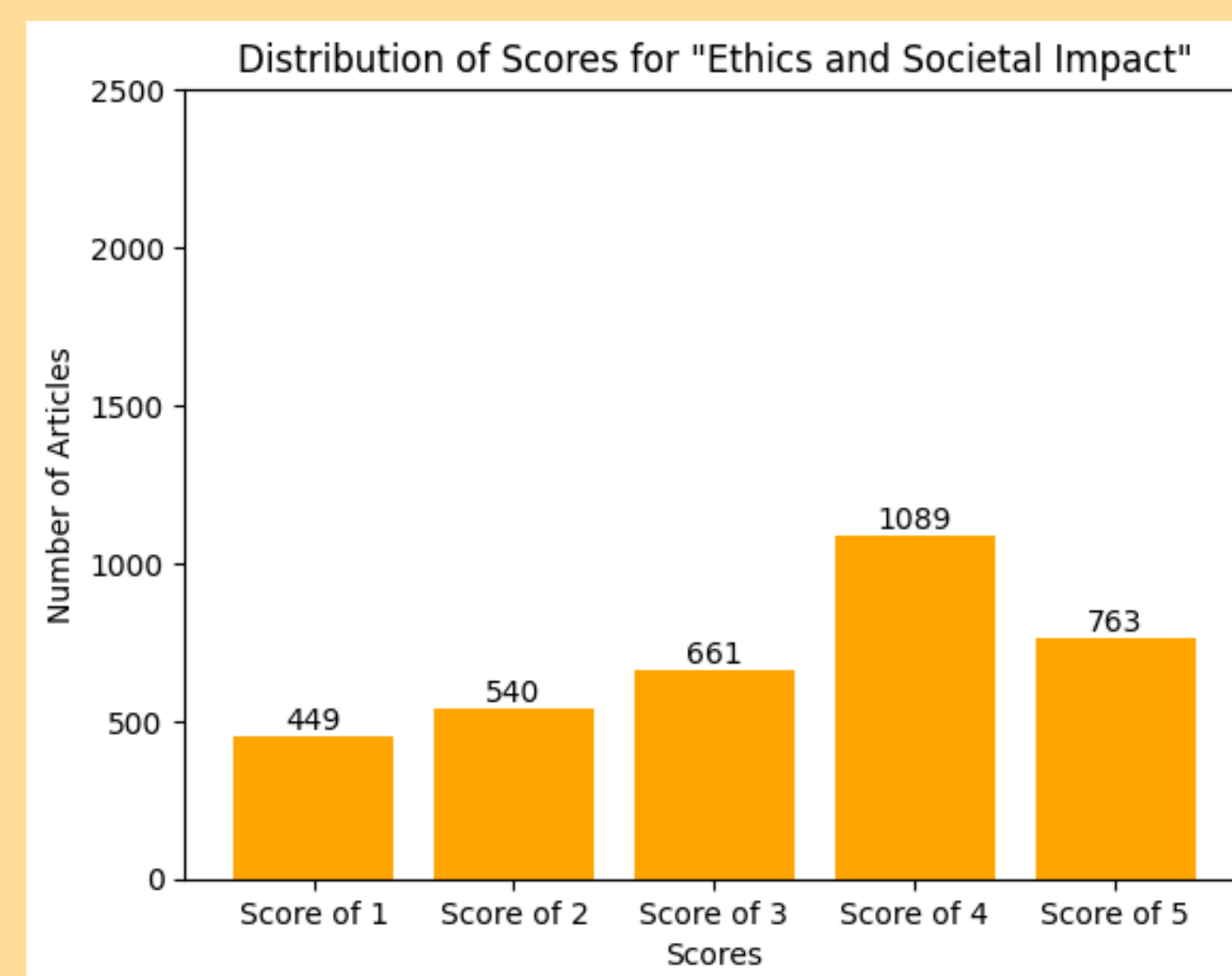


Figure 3. Bargraph for the distribution of scores for "ethics and societal impact"

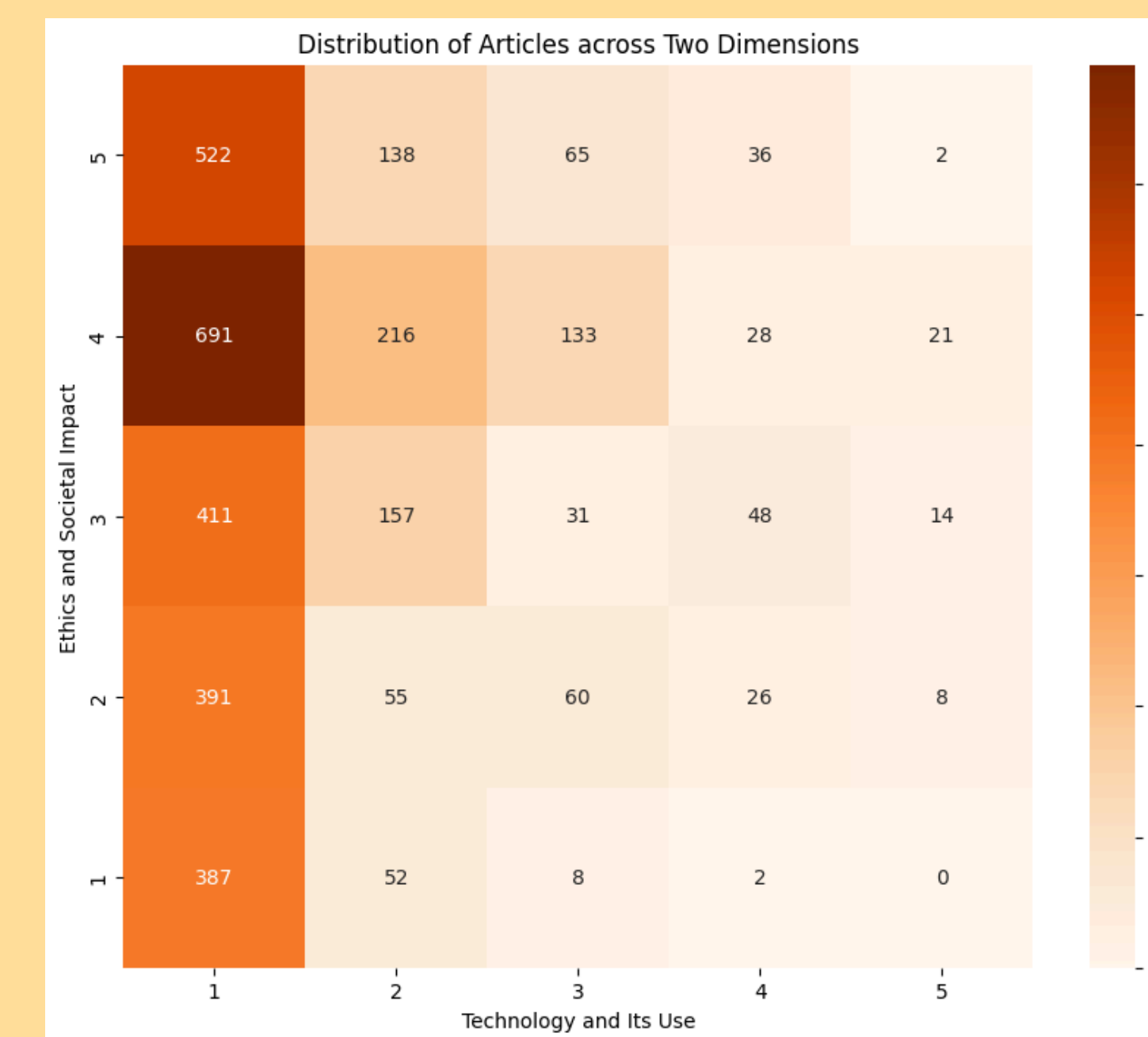


Figure 4. Heatmap for the distribution of articles across the two dimensions "technology and its use" and "ethics and societal impact"

SideChat

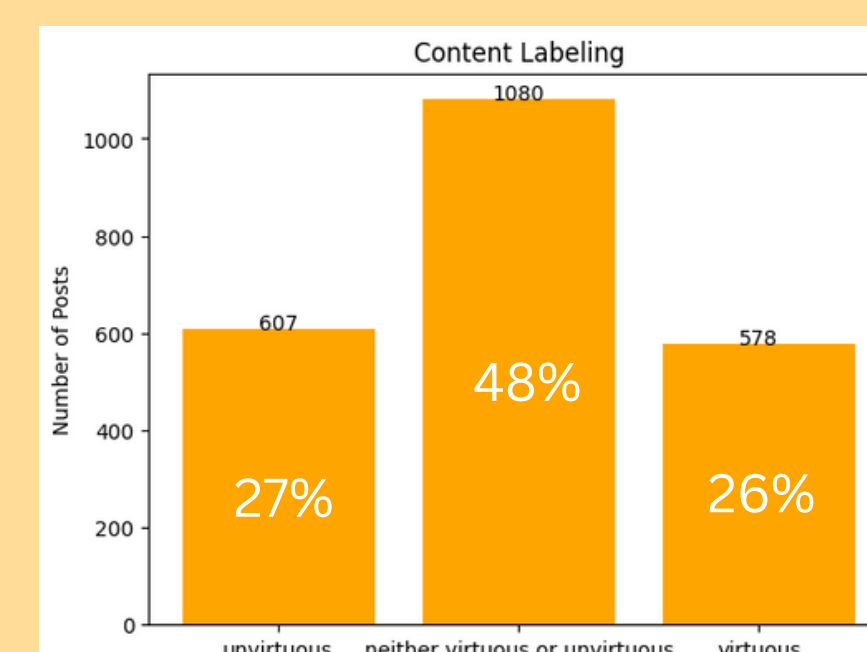


Figure 5, Bar graph showing how gpt labeled the content of posts within the sample set of 2265 posts

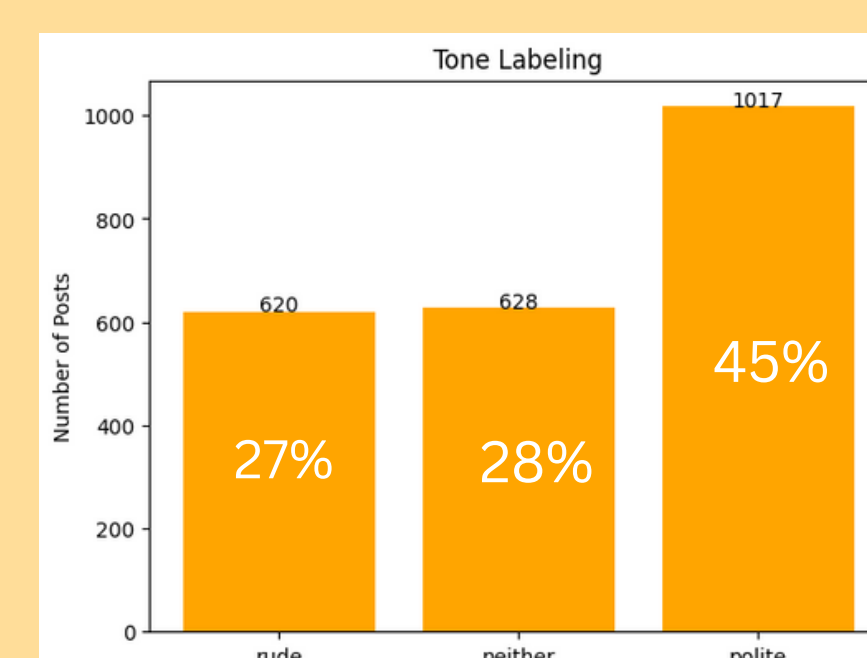


Figure 6. Bar graph showing how gpt labeled the tone of posts within the sample set of 2265 posts

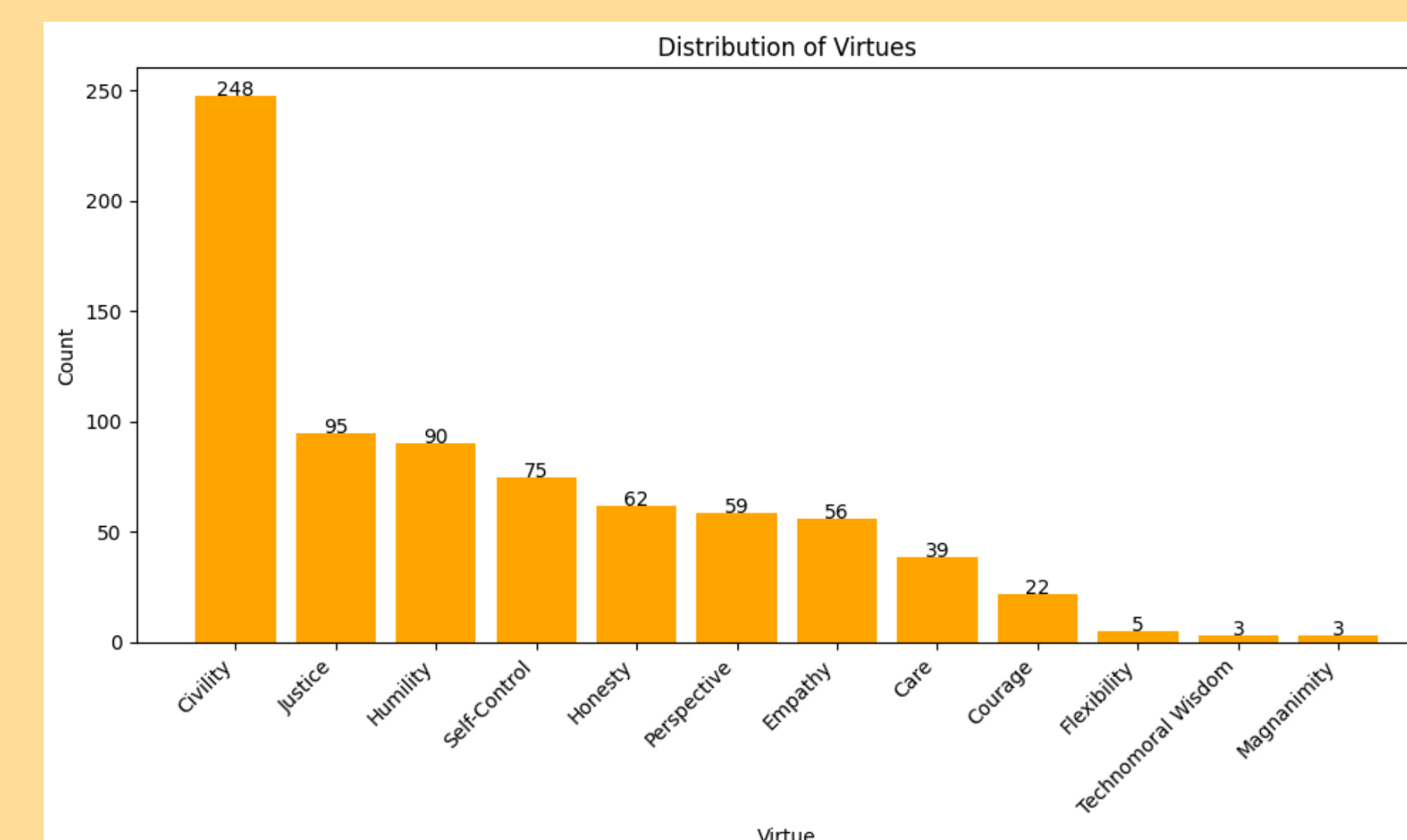


Figure 7. Bar graph of the distribution of the strongest virtue present in each post in a sample set of 2265 posts

Labeled Example:

Message: El Table workers are the best 🙌🙌🙌🙌

Tone: 1
Content: 1
Strongest Virtue: Civility

Tone Score Reasoning:
The message is very positive and uplifting, using praise and friendly emojis.

Content Score Reasoning:
It promotes a positive social interaction by showing appreciation and respect for others' work.

Discussion

Course Catalog: The Wellesley course curriculum displays many virtues, particularly perspective, justice, and empathy. Very few courses have no significant emphasis on at least one virtue, with the vast majority of courses somewhat focusing on at least one technomoral virtue (scores of 3 and 4). **65 courses** (3%) exhibited at least 4 virtues, with **17 (0.93)** courses exhibiting at least 5.

Wellesley News: Having a score of 4 or 5 was defined as having high relevance to the dimension the score measures. 5.28% (185/3504) of the articles collected received a score of high relevance to technology and its use and 52.85% (1852/3504) of the articles collected received a score of high relevance for ethics and societal. Only 2.48% (87/3504) of the articles received a score of high relevance for both dimensions, meaning that only a small number of them extensively discuss ethics and societal impact under technological contexts.

Sidechat: The **3 most prevalent virtues** were **civility** with 248 (11%) posts, **justice** with 95 (4%) posts, and **humility** with 90 (4%) posts. The number of virtuous posts GPT labeled is quite surprising. Despite the rhetoric that Sidechat is a toxic place, the data shows less than 30% of posts were labeled as unvirtuous or uncivil.

Overall, there is a presence of virtues within the three datasets was a pleasant surprise. If these virtues are to be enforced it could strengthen the relationship in the way individuals use technology, which Vallor's emphasizes.

Future Work

Course Catalog

- Interview students/professors about courses that focus on the virtues
- Consider how curricular and extracurricular activities provide opportunities for practicing these virtues
- Build an interactive website to provide an annotated exploration of the curriculum across the virtues

Wellesley News

- Include articles before 2013 and after 2022 into the dataset to analyze its changes regarding technological and ethical topics over time
- Encourage CS students to contribute news articles to Wellesley News that cover topics of ethics of technology and its dilemmas

Sidechat

- **Interview students** about their opinons on the influence of Sidechat conversations on their ethical decision-making
- Label the virtuous posts **based on topic** (ex: academics, politics, relationships, ect.)

Acknowledgement

Thank you to Prof. Eni Mustafaraj and team leader Johanna Lee for their care, support, and most of all, patience. Thank you National Science Foundation and T.T. and W.F. Chao Summer Scholars Program in Natural Science for funding our research.

An Assessment of GPT-Prompting for Labeling Text in the Context of Virtue Ethics



Evelin Buestan, '26, Lucy Engels, '27, Artemist Pierre '27, Yuhe Xue '27
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Background

- This poster explores using GPT prompting to label texts of three contextual datasets that are analyzed in the companion poster, *A Computational Examination of Virtue Ethics Presence in Three Textual Datasets*.
- Due to the large number of textual units in each dataset, **manual labeling would have been costly and time consuming**. Applying GPT made what would have taken days of work of manual labeling be completed in a much more efficient way, within hours. However, finding the right prompt is a laborious process.
- GPT Prompting** refers to interacting with a Generative Pre-trained Transformer (GPT) by **providing it with specified instructions** to perform certain tasks and return desired responses.
- Model GPT - 4o** was used to label text in this study.

How to Measure Text

To compare text units, we use GPT to assign ordinal labels (scores) according to various scales, as described below.

Wellesley Course Catalog: GPT is prompted to give a score on a 5-point scale evaluating the extent to which each of Shannon Vallor’s techno moral virtues is present in the text. A score of **1** indicates that the **virtue is minimally present** in the course and **5** indicates that it is **very present**.

Wellesley News: A 5-point scale was used to measure the relevance of each article to two dimensions, “technology and its use” and “ethics and societal impact”. A score of **1** means “**minimally mentions**”, and a score of **5** means “**heavily mentions**”. A score of 4 or 5 was then defined as an indication of the high relevance of the article to the dimension the score measures

Sidechat: A 3-point scale was used to measure two dimensions, **Tone** and **Content**. The Tone is scored from **-1 to 1**, where -1 means being **rude** and 1 means being **polite**. The Content also use the same scale, where -1 means the **opposite of Shannon Vallor’s technomoral virtues**, and 1 means **displaying the virtue fully**.

Evaluating GPT Results

Validation Test

The Cohen’s Kappa is used to measure the agreement between the results of GPT labeling and Human labeling.

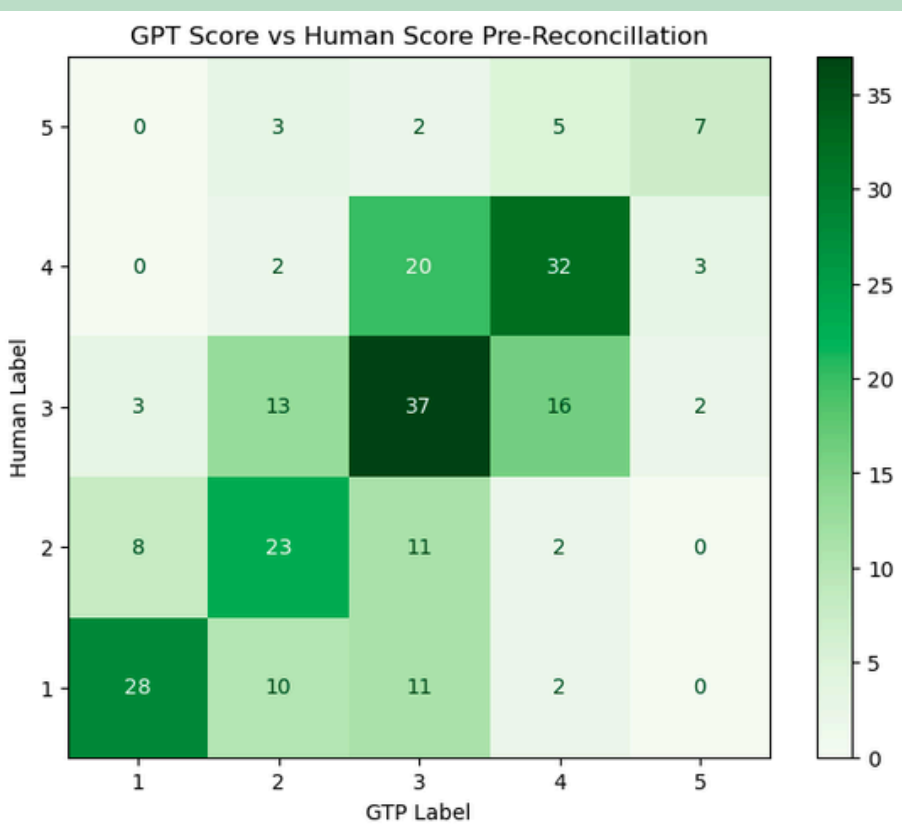
Cohen’s Kappa (K)

$$K = \frac{(\text{observed agreement} - \text{expected agreement})}{(1 - \text{expected agreement})}$$

- $\kappa < 0$: Less than chance agreement
- $\kappa = 0 - 0.20$: Slight agreement
- $\kappa = 0.21 - 0.40$: Fair agreement
- $\kappa = 0.41 - 0.60$: Moderate agreement
- $\kappa = 0.61 - 0.80$: Substantial agreement
- $\kappa = 0.81 - 1.00$: Almost perfect agreement

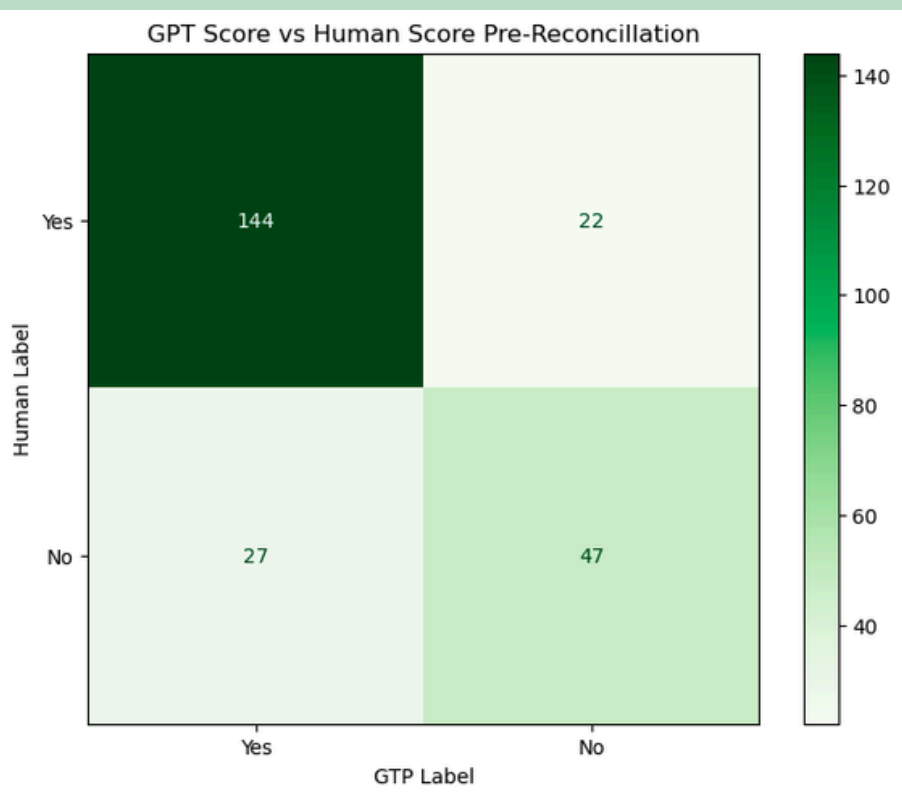
Course Catalog

- Used GPT to examine how much a course description focused on each of Shannon Vallor’s technomoral virtues
- It takes **2-3 minutes for a human** to evaluate a single course without a justification, and **3-5 minutes** with a justification
- Takes **15-20 seconds for AI** to label a course with justifications
- Prompt provided a rubric to use when scoring and two examples, using the “few shot” technique of prompting
- Used a **sample of 20 courses** to evaluate agreement



0.38

5-point agreement
52% (127 / 240)
exact agreement
89% (213/240)
within 1 point



0.51

80% (191 / 240)
Binary agreement

GPT Example: Disagree

Course: CPLT 294: Utopia and Dystopia in Literature

Virtue: Civility **GPT Ranking:** 3 **Human Ranking:** 5

GPT Justification:

Discussions around utopian and dystopian societies often involve collective deliberation about societal norms and responsibilities, promoting civility and respect.

GPT Example: Agree

Course: ARTS 321/ CAMS 321 - Advanced New Media

Virtue: Empathy **GPT Ranking:** 4 **Human Ranking:** 4

GPT Justification:

Collaborative projects and peer critiques in the course promote empathy, as understanding others' perspectives is necessary for effective collaboration and constructive feedback.

Discussion

Course Catalog:

- While most GPT justifications lined up with a human’s understanding, there were some places where a human’s interpretation significantly differed from GPT’s understanding
- Creating a successful prompt required many iterations. Reaching a Cohen’s Kappa score of 0.38 required 14 attempts of iterating across the prompt and the rubric
- There is a level of subjectivity, as illustrated in the example where GPT and a human rater disagreed.

Wellesley News:

- GPT tends to generate more accurate scores when the prompt asks it to include justifications for its scores.
- GPT also generates responses with lower variability when the prompt specifies a narrow range of options for the desired response. For example, question “answer whether the article is subjective or objective” gives either “subjective” or “objective” as the answer, while question “answer whether the article is subjective or objective” may lead to various unexpected answers such as “NA” or “objective and neutral”.
- Clear and concise prompts make GPT generate best responses, yet what is clear and concise enough to GPT could only be realized through trial and error.

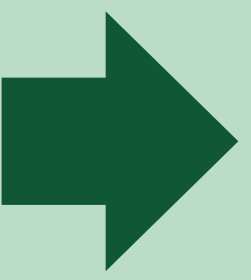
Sidechat:

- GPT scored the **contents** of **34** posts as unvirtuous while the human labeler scored **13** as such.
- GPT is more likely than the individual to score a post as unvirtuous.
- This is partially due to GPT’s **misinterpretations of emojis and slang**.
- The Cohen’s Kappas however show substantial agreement so using **GPT is a dependable choice to label Sidechat posts** and will be improved on.

Wellesley News

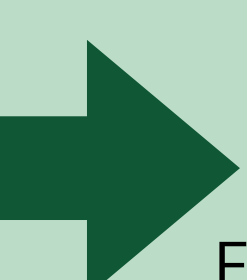
Prompt for GPT

```
message:
  • [{"role":"system","content":"You are a helpful assistant trained to give an answer to user's question. You will format the answer as a JSON object with keys Technology and Its Use, Ethics and Societal Impact, Subjective or Objective, Stance, Summary, and Reason for the Scores. Each JSON object should not be in a docstring"},
  • {"role":"user","content":"I need your help with summarizing a news article with respect to a few dimensions:
    ◦ to what extent it is about technology and its use (respond with a score from a 5-point scale from 1, representing not at all, to 5, representing entirely)
    ◦ to what extent it is about ethics or societal impact (respond with a score from a 5-point scale from 1, representing not at all, to 5, representing entirely)
    ◦ whether it's subjective or objective (respond one word out of subjective or objective)
    ◦ what is the stance of the author (if the article is subjective, respond one word out of approving, disapproving, or neutral; if the article is objective, respond NA)
    ◦ then provide a very short summary of the article (Respond to this in one string)
    ◦ then provide you reasoning for the scores you gave (Respond to this in one string). End of instructions. What follows is the text of the article:(text_to_analyze)"}]]
```



An example of an output from GPT that received 5 for both dimensions

original_articles	response
<pre>{ • "title": "Scientists Construct an Embryo Derived from Three Parents", • "authors": xxx • "date": : xxx • "category": : xxx • "url": : xxx • "body": : xxx }</pre>	<pre>{ • "Technology and Its Use": 5, • "Ethics and Societal Impact": 5, • "Subjective or Objective": "objective", • "Stance": "NA", • "Summary": :xxxx • "Reason for the Scores": : xxx }</pre>



Validation test

- Human readers manually labeled a sample of 30 articles
- Cohen’s Kappa (k)

For “Technology and Its Use”:

0.491

For “Ethics and Societal Impact”:

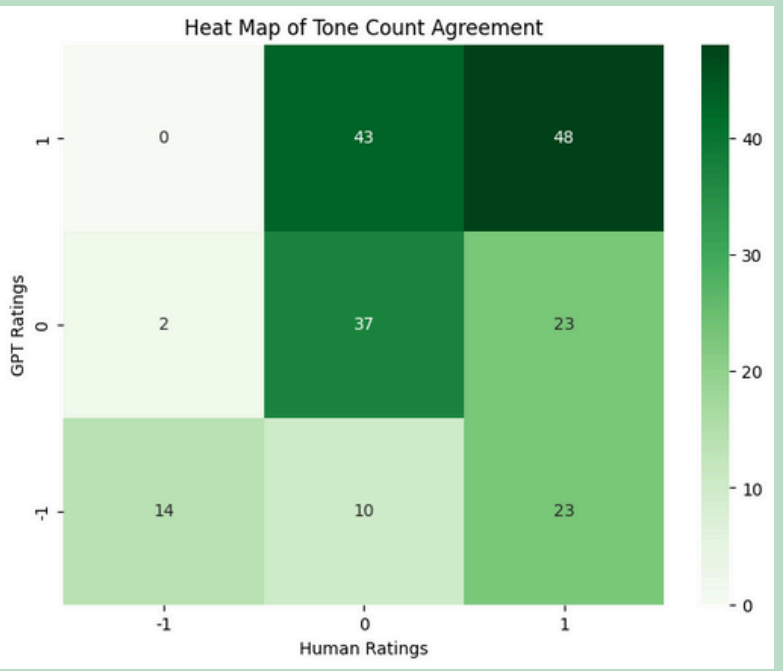
0.747



Conclusion: While GPT can be trusted to perform text labeling on Wellesley News, prompt could be further modified to enhance GPT’s agreements with human readers on scoring for “Technology and Its Use”

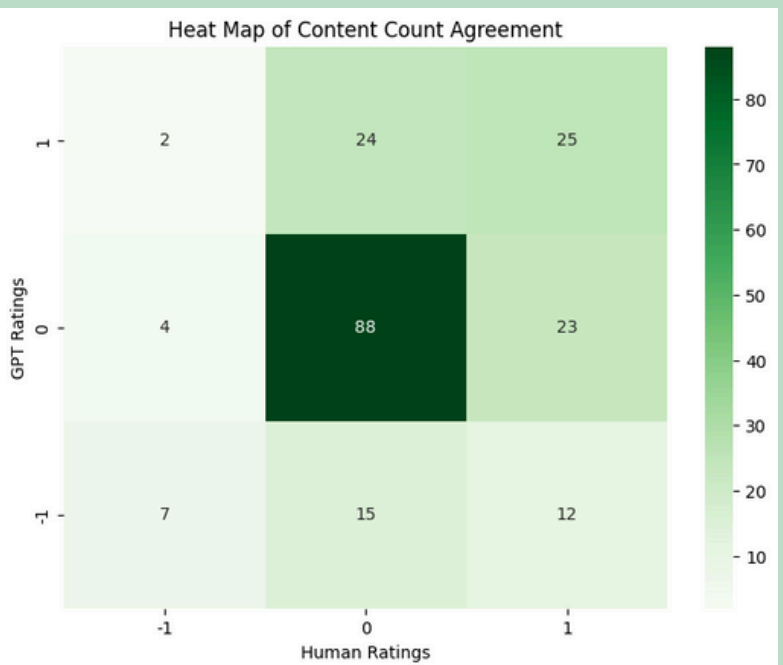
Sidechat

- Labeling 200 posts took a human **5 hours** while GPT took **2 minutes**.
- Kappa for **Tone** and **Content** show **significant agreement**
- Agreement for **unvirtuous** tone and content is **comparatively lower** than virtuous and neutral agreement
- GPT can **misinterpret** emojis and slang



0.76

Tone Agreement

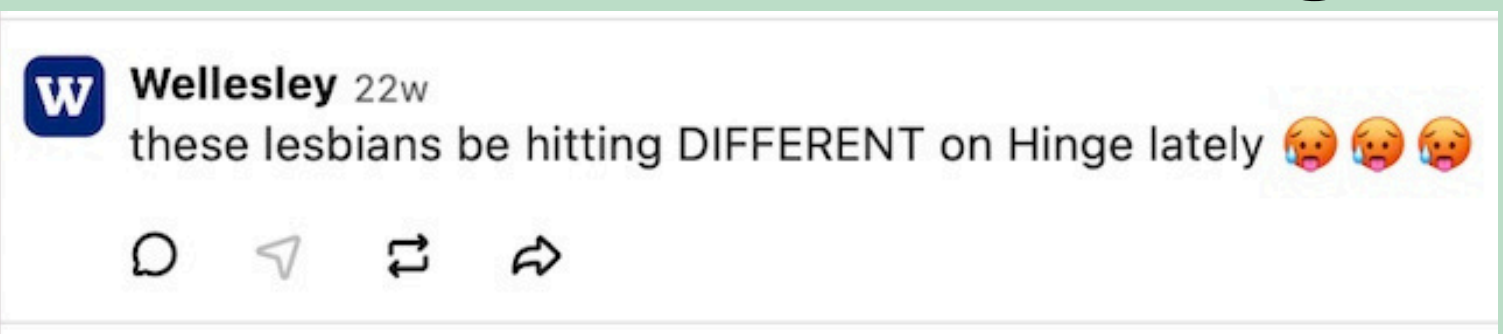


0.73

Content Agreement

Figure 2: Heat Map of the Tone and Content Agreement for 200 messages where -1 represents vices, 0 no virtue or vice present, and 1 represents virtues.

GPT’s Inaccurate Scoring



Tone: -1
Content: -1

Tone Score Reasoning: The tone is rude and dismissive, indicated by the use of negative emojis and an overall disrespectful attitude.

Content Score Reasoning: The message does not address any virtues and contains a general sense of negativity and perhaps implicit bias or judgment.

How would you score this?

Future Work

For all three datasets, we aim to:

- Improve GPT prompting by making the prompts more clear and concise in order for the responses to reach better agreement with human readers
- Take a step forward to explore GPT few-shot learning (providing a few labeled examples) or fine tuning (training with many labeled examples)
- Validate results with more than one human reader to see the inherent level of subjectivity of the tasks

Acknowledgement

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